5. Project Development Phase

Implementation of backend, AI modules and frontend

Backend Development

FastAPI is used as the backend framework. Each educational capability is implemented as a separate module and exposed through a RESTful endpoint.

Explanation Module

The documented implementation uses LaMini-Flan-T5 to provide concise, context-aware explanations that break complex topics into simpler language.

Q&A; Module

Gemini 1.5 Pro is documented as the model used for general academic and knowledge-based question answering.

Quiz Module

The module asks Gemini to generate three MCQs with four options each. The documented implementation cleans Markdown code blocks before parsing the response as JSON and returns an error message if generation or parsing fails.

Summary Module

Gemini is used to condense long paragraphs into concise, understandable summaries while retaining core information.

Learning Path Module

Gemini generates a structured learning path from beginner to advanced topics, with suggested resources and guidance.

Frontend Integration

The HTML form allows users to select the task and enter content. The form sends a POST request to the FastAPI backend and shows the result in the interface.

Local Execution

The project document specifies running the application with: uvicorn main:app --reload.